MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS) SILICON

TYPES: 2N6605, 2N6606, 2N6607
2N6608, Non-TX, TX, and TXV

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNPN silicon low power, reverse-blocking-triode, thyristors, and is in accordance with MIL-S-19500 except as otherwise specified herein. The prefix "TX" is used on devices submitted to and passing the special process-conditioning, testing, and screening as specified in 4.6. The prefix "TXV" is used on devices submitted to and passing the inspection specified in 4.5 and 4.6.

1.2 Physical Dimensions. See Figure 1 (TO-18)

1.3 Maximum Ratings. (See 6.3)

	V_R	V_D	I_O	i_{TSM}(surge) 1 cycle 2/	V_{KGM}	Top and Tstg
	Vdc	V(pk)	A	a	V(pk)	°C
2N6605	30	30 1/	0.35	6	8	-65° to +150
2N6606	60	60 1/	0.35	6	8	-65 to +150
2N6607	100	100 1/	0.35	6	8	-65 to +150
2N6608	200	200 1/	0.35	6	8	-65 to +150

1/ Gate connected to cathode through 1000-ohm resistor.

2/ Surge current is non-recurrent. The rate of rise of peak surge current shall not exceed 40-amperes during the first 5 μseconds after switching from the "off" (blocking) to the "on" (conducting) state. This time is measured from the point where the thyristor voltage has decayed to 90-percent of its initial blocking value.

1.4 Primary Electrical Characteristics.

	V_T	$I_{H X}$	V_{GT} 1/ I_{GT} 1/ T_A= -65°C to +150°C		I_{DX} V_D=Rated	I_R V_R=Rated
	v	mAdc	Vdc	μAdc	μAdc	μAdc
Min	---	---	0.2	---	---	---
Max	1.7	5.0	0.8	200	0.12	0.25

1/ Gate connected to cathode through 1000-ohm resistor.

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

MIL-F-14256 - Flux, Soldering, Liquid, Rosin Base.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-750 - Test Methods for Semiconductor Devices

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements for the thyristors shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500 and as follows:

R_{GK} --- External resistance between gate and cathode terminals.

V_{AA} --- Anode power supply voltage (d.c.).

3.3 Design, Construction, and Physical Dimensions. Thyristors shall be of the design, construction, and physical dimensions shown on Figure 1.

3.3.1 Lead Material and Finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be Gold- or Tin-plated. Where a choice of lead material and finish is desired, it shall be specified in the contract or order (see 6.2).

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3.4 Performance Characteristics. Performance characteristics shall be as specified in Tables I, II and III, and as follows:

3.4.1 Process-Conditioning, Testing, and Screening for "TXV" and "TX" Types. Process-conditioning, testing, and screening for the "TXV" types shall be as specified in 4.5 and 4.6, and "TX" types shall be as specified in 4.6.

3.4.2 Internal Visual (PRECAP) Inspection and Process-Conditioning, Testing and Screening of "TXV" Types. The "TXV" device type shall in addition to all performance requirements, be internally visually inspected and process-conditioned tested, and screened in accordance with 4.5.1.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the thyristor at the option of the manufacturer:

(a) Country of origin

(b) Manufacturer's identification

(c) Polarity

3.5.1 "TXV" Marking. Devices in accordance with the "TXV" requirements shall be marked with "TXV" immediately following the JAN prefix.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and Inspection. - Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein. - The lot accumulation period requirements shall be 3 months in lieu of 6 weeks. If a lot is made up of a collection of sublots, Subgroups 3 and 4 of Group B inspection and Subgroups 1,2,3,4 and 5 of Group C inspection may be performed on an inspection sublot of any type, subgroups 1 and 2 of Group B inspection may be performed on a full sample of each sublot present in the lot or single sample consisting of an equal number of devices from each sublot and Subgroups 5 and 6 or Group B inspection and Subgroups 6 and 7 of Group C inspection may be performed in accordance with Paragraph 40.1 of MIL-S-19500 or performed on a full sample size of the prime device type only. (The prime device type shall be determined by the following descending order 2N6608, 2N6607, 2N6606 and 2N6605.)

4.2 Qualification Inspection. Qualification inspection shall consist of the examinations and tests specified in Tables I, II, and III. The requirement for tightened LTPD for each subgroup, as required in MIL-S-19500, Appendix C, paragraph 50.2 shall not apply.

4.2.1 Qualification Testing. - Any level, JAN, JANTX or JANTXV may be used for qualification testing; however, the test report shall contain only that data specified in 4.2. Upon request to the qualifying activity, qualification will be extended to include the other device levels.

4.2.2 Variables Data Criteria For Qualification Inspection. For qualification inspection, Group A variables data shall be submitted on a sample

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4.2 Qualification Inspection (continued).

4.2.2 Variables Data Criteria For Qualification Inspection (continued). associated with the lowest LTPD specified in Table I. However, in subgroups with higher LTPD's, acceptance criteria shall be based on a pre-designated portion of that sample.
In addition, variables data shall also be submitted on the Group B and C pre-test and post-test endpoints as specified in Table II. The remaining Group B and C samples need not be submitted to all tests of Group A.

4.3 Quality Conformance Inspection. Quality conformance inspection shall consist of Group A, B, and C inspections. When specified in the contract or order, one copy of the quality conformance inspection data, pertinent to the device inspection lot, shall be supplied with each shipment by the device manufacturer (see 6.2). Quality Conformance inspection shall include inspection preparation for delivery (see 5.1 herein).

4.3.1 Group A Inspection. Group A inspection shall consist of the examinations and tests specified in Table I.

4.3.2 Group B Inspection. Group B inspection shall consist of the examinations and tests specified in Table II.

4.3.3 Group C Inspection. Group C inspection shall consist of the examinations and tests specified in Table III. This inspection shall be conducted on the initial lot and thereafter every 12 months during production.

4.3.4 Group B and Group C Life Test Samples. Samples that have been subjected to Group B, 170 hours life-test, may be continued on test to 500 hours in order to satisfy Group C life-test requirements. These samples shall be pre-designated, and shall remain subjected to the Group C, 500 hour acceptance evaluation after they have passed the Group B, 170 hour acceptance criteria. The cumulative total of failures found during 170 hour test and during the subsequent interval up to 500 hours shall be computed for 500 hour acceptance criteria (see 4.3.3).

4.4 Methods of Examination and Test. Methods of examination and test shall be as specified in Table I, II, and III and as follows:

4.4.1 Resistance to Solvents. Thyristors shall be subjected to tests in accordance with methods 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the thyristor body where marking has been applied shall be brushed.

(b) After subjection to the tests there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

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4.4 Methods of Examination and Test. (continued)

4.4.2 Solderability Test. The following exceptions shall apply to this test:

(a) Use of RMA flux per MIL-F-14256.

(b) Solder temperature 230°C to 260°C.

(c) Terminal coverage, 90 percent.

4.5 Process-Conditioning, Testing, and Screening for "TXV" Types. Process-conditioning, testing, and screening for "TXV" types shall be in accordance with 4.5.1 through 4.6.8.1 and Figure 3.

4.5.1 Internal Visual (PRECAP) Inspection and Process-conditioning, Testing, and Screening For "TXV" Types. The internal visual inspection shall be performed in accordance with test method 2072 of MIL-STD-750 prior to encapsulation on a 100 percent basis and shall also include process-conditioning, testing, and screening as specified in 4.6. The manufacturer shall permit the authorized government representative to witness concurrent with time of manufacturer's performance of these tests, the process-conditioning, testing, and screening of devices. Those conditioning and screening tests normally performed by a manufacturer as standard production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than the test specified herein.

4.6 Process-Conditioning, Testing, and Screening for "TXV" and "TX" Types. The procedure for process-conditioning, testing, and screening for "TXV" and "TX" types shall be in accordance with MIL-S-19500 and 4.6.1 through 4.6.8.1 and Figure 3. At the option of the manufacturer, "TX" devices may be processed in accordance with Figure 4 and 4.6.1 through 4.7.

4.6.1 Quality Assurance (Lot Verification). Quality assurance shall be in accordance with MIL-S-19500 except lot records shall be kept for a 2 year minimum.

4.6.2 High-Temperature Storage. All devices shall be stored for at least 48 hours at a minimum temperature (T_A) of 150°C.

4.6.3 Thermal Shock (Temperature Cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with MIL-STD-750, Method 1051, test condition F, except that 10 cycles shall be continuously performed and the time at the temperature extremes shall be 15 minutes minimum.

4.6.4 Acceleration. All devices shall be subjected to acceleration test in accordance with MIL-STD-750, Method 2006, with the following exceptions: the test shall be performed one time in the Y_1 orientation only at a peak level of 20000G minimum. The one-minute hold-time requirement shall not apply. (This test applies only when bonding wire other than aluminum is used.)

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4.6 Process-Conditioning, Testing, and Screening for "TXV" and "TX" Types. (continued)

4.6.5 Hermetic Seal Tests. All devices shall be subjected to hermetic seal tests (fine-leak followed by gross-leak) with test conditions as specified in 4.6.5.1 and 4.6.5.2. Failed devices from either test shall be removed from the lot.

4.6.5.1 Hermetic Seal (Fine-Leak) Test. All devices shall be fine-leak tested in accordance with MIL-STD-750, Method 1071, test condition G or H. The leak-rate rejection criterion shall be 1 X 10^7 cubic centimeters of helium per second when measured at a differential pressure of one atmosphere.

4.6.5.2 Hermetic Seal (Gross-Leak) Test. All devices shall be tested for gross-leaks in accordance with MIL-STD-750, Method 1071, test conditions, A, C, D, or F.

4.6.6 Pre-Burn-In Tests. The parameters of Table IV shall be measured for all devices in the lot. All devices which fail to meet these requirements shall be removed from the lot and the quantity removed shall be noted on the lot history.

4.6.7 Burn-In Test. All devices shall be operated for 72 hours minimum under the following conditions:

T_A = 125°C

2N6605 = V_D = 30Vdc
2N6606 = V_D = 60 Vdc
2N6607 = V_D = 100 Vdc
2N6608 = V_D = 200 Vdc

R_{GK} = 1000 ohms

All devices which turn "on" during the burn-in test shall be removed from the lot and the quantity removed shall be noted on the lot record.

4.6.8 Postburn-In Tests. All devices in the lot shall be tested to all the parameters of Table IV after burn-in.

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TABLE I. Group A Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 1			10	10				
Visual and mechanical inspection	2071							
Subgroup 2			5	2				
Reverse blocking current	4211	DC method, bias cond.D			I_R			
2N6605		V_R = 30 Vdc				---	0.25	µAdc
2N6606		V_R = 60Vdc				---	0.25	µAdc
2N6607		V_R = 100Vdc				---	0.25	µAdc
2N6608		V_R = 200Vdc				---	0.25	µAdc
Forward blocking current	4206	DC method, bias cond.B			I_{DX}			
		R_2 = 1000 ohms						
2N6605		V_D = 30Vdc				---	0.12	µAdc
2N6606		V_D = 60Vdc				---	0.12	µAdc
2N6607		V_D = 100Vdc				---	0.12	µAdc
2N6608		V_D = 200Vdc				---	0.12	µAdc
Subgroup 3			5	3				
Reverse blocking current	4211	DC method, bias cond.D;			I_R			
		T_A = 150°C						
2N6605		V_R = 30Vdc				---	100	µAdc
2N6606		V_R = 60Vdc				---	100	µAdc
2N6607		V_R = 100Vdc				---	100	µAdc
2N6608		V_R = 200Vdc				---	100	µAdc

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TABLE I. Group A Inspection (continued)

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 3 (cont.)								
Forward blocking current	4206	DC method, bias cond. B			I_{DX}			
		T_A = 150°C						
		R_2 = 1000ohms						
2N6605		V_D = 30Vdc				---	20	µAdc
2N6606		V_D = 60Vdc				---	20	µAdc
2N6607		V_D = 100Vdc				---	20	µAdc
2N6608		V_D = 200Vdc				---	20	µAdc
Gate trigger voltage	4221	T_A = 150°C;			V_{GT}			
		R_L = 100ohms						
		R_e = 1000ohms						
2N6605		$V_2=V_D$= 30Vdc				0.2	---	Vdc
2N6606		$V_2=V_D$= 60Vdc				0.2	---	Vdc
2N6607		$V_2=V_D$= 100Vdc				0.2	---	Vdc
2N6608		$V_2=V_D$= 200Vdc				0.2	---	Vdc
Subgroup 4								
Gate trigger voltage and current	4221	$V_2=V_D$=6Vdc						
		R_L=100 ohms						
		R_e=1000 ohms			V_{GT}	---	0.80	Vdc
					I_{GT}	---	200	µAdc
Forward "on"voltage	4226	i_D=2a(pk) (pulse;pulse width=2msec, max; duty cycle=2%max			v_T	---	1.7	v(pk)

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TABLE I. Group A Inspection (continued)

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 4 continued								
Holding current	4201	Bias cond. B; V_{AA}=24 Vdc max; I_{F1}=100 mAdc; I_{F2}=10 mAdc; gate trigger source volt= 6Vdc; trigger pulse width=25µsec min: R_2=330 ohms R_3=1000ohms			$I_{H X}$	---	5.0	mAdc

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TABLE II. Group B Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 1			10	10				
Reverse gate current	4219	V_{KG}=6Vdc			I_{KG}	---	0.1	µAdc
Surge current	4066	T_A=25°C; i_{TSM}(surge)= 6a,10 surges at 1 per min.; I_O=0 at rated V_{RM}=0; f=60Hz			---	---	---	---
Exponential rate of voltage rise	4231	Bias cond. A; T_A= 70 °C; repetition rate=60pps; 50Ω≤R_L≤400Ω; C=.1to 1ufd; test duration = 15 sec; dv/dt=200V/µsec; R_e=1000 ohms V_{GG}=-2V			V_{DX}			
2N6605		V_{AA}= 30Vdc				27	---	Vdc
2N6606		V_{AA}= 60Vdc				57	---	Vdc
2N6607		V_{AA}= 100Vdc				95	---	Vdc
2N6608		V_{AA}= 200Vdc				190	---	Vdc
Endpoints:								
Reverse blocking current	4211	DC method, bias cond.D			I_R			
2N6605		V_R= 30Vdc				---	0.25	µAdc
2N6606		V_R= 60Vdc				---	0.25	µAdc
2N6607		V_R= 100Vdc				---	0.25	µAdc
2N6608		V_R= 200Vdc				---	0.25	µAdc

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TABLE II. Group B Inspection (continued)

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 1 (cont.)								
Endpoints: cont.								
Forward blocking current	4206	DC method, bias cond.B R_2 = 1000ohms			I_{DX}			
2N6605		V_D= 30Vdc				---	0.12	µAdc
2N6606		V_D= 60Vdc				---	0.12	µAdc
2N6607		V_D= 100Vdc				---	0.12	µAdc
2N6608		V_D= 200Vdc				---	0.12	µAdc
Gate trigger voltage and current	4221	V_2=V_D=6Vdc; R_L = 100 ohms						
		R_e =1000 ohms			V_{GT}	---	0.80	Vdc
					I_{GT}	---	200	µAdc
Forward "on" voltage	4226	i_D=2a(pk) (pulse); pulse width=2 msec max; duty cycle=2% max			v_T	---	1.7	v(pk)
Subgroup 2			10	10				
Low temperature tests:		T_A= -65°C						
Gate trigger voltage and current	4221	V_2=V_D=6Vdc; RL=100 ohms			V_{GT}	---	0.9	Vdc
		R_e=1000 ohms			I_{GT}	---	700	µAdc
Forward "on" voltage	4226	i_D=2a(pk) (pulse);pulse width=2msec max; duty cycle= 2% max			v_T	---	1.85	v(pk)

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TABLE II. Group B Inspection (continued)

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 2(cont.)								
Holding current	4201	Bias cond.B; V_{AA}= 24Vdc max; I_{F1}=100 mAdc; I_{F2}=10 mAdc; gate trigger source voltage=6Vdc; trigger pulse width=25μsec min; R_2=330 ohms; R_3=1000 ohms			I_{H} X		8.0	mAdc
Subgroup 3			10	10				
Solderability	2026	(see 4.4.2)			---	---	---	---
Thermal shock (temperature cycle)	1051	Test cond.F 10 cycles:time at temperature extremes =15 minutes min.			---	---	---	---
Thermal shock (glass strain)	1056	Test cond.A			---	---	---	---
Hermetic seal	1071	Test cond. G or H for fine leaks; test cond.A,B,C,D, or F for gross leaks			---		$5X10^{-8}$ atm cc/sec	
Moisture resistance	1021				---	---	---	---
Endpoints: (same as subgroup 1)								

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TABLE II. Group B Inspection (continued)

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 4			10	10				
Shock	2016	Nonoperating; 1500 G, 0.5 msec, 5 blows each in orientations; X_1, Y_1, Y_2 and Z_1			---	---	---	---
Vibration variable frequency	2056	Nonoperating			---	---	---	---
Constant accelerations	2006	20,000 G; in orientations X_1, Y_1, Y_2 and Z_1			---	---	---	---
Endpoints: (same as subgroup 1)								
Subgroup 5			7	7				
High temperature life (non-operating)	1032	T_A=150°C t=170 hrs (see 4.3.4)						
Endpoints: (same as subgroup 1)								
Subgroup 6			7	7				
Steady state Operation Life	1027	T_A= 25°C t=170 hours (see 4.3.4 and figure 5) R_{GK}=1000 ohms						

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TABLE II. Group B Inspection (continued)

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 6 (cont.)								
Steady State Operation Life (continued)	1027	I_O = 200mAdc						
2N6605		V_D = 30Vdc						
2N6606		V_D = 60Vdc						
2N6607		V_D = 100Vdc						
2N6608		V_D = 200Vdc						
Endpoints:								
(same as subgroup 1)								

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TABLE III. Group C Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 1			20	20				
Physical dimensions	2066	(see figure 1			---	---	---	---
Subgroup 2			15	15				
Resistance to solvents		Method 215 of MIL-STD-202 (See 4.4.1) Electrical rejects may be used						
Subgroup 3			15	15				
Salt atmosphere (corrosion)	1041	(Electrical rejects may be used)			---	---	---	---
Subgroup 4			15	15				
Terminal strength (lead fatigue)	2036	Test cond. E (electrical rejects may be used)						
Subgroup 5			10	10				
Pulse response: Turn-on time	4223	t_{P1}=10-30 usec; V_{GG}=15V; R_e=2000 ohms;			t_{on}	0.3		us
2N6605		V_{AA}=30Vdc R_L=2 ohms						
2N6606		V_{AA}=60 Vdc R_L=4 ohms						
2N6607		V_{AA}=60 Vdc R_L=4 ohms						
2N6608		V_{AA}=60 Vdc R_L=4 ohms						

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TABLE III. Group C Inspection (continued)

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Subgroup 5 (cont.)								
Gate turn-off	4225	Tc=70°C; switching rate = 1Hz; on-state current duration=.5 to 6 ms; I_{GT}=.5-1mA for 10-100μsec;			t_{off}			
Test #1		On state current=800mA I_{GQ1}=80 mA					Device must not turn-off	
Test #2		On state current=2000 mA; I_{GQ2}=200 mA					Device must not turn-off	
Subgroup 6			7	7				
High temperature life	1032	T_A=150°C t=500 hours (see 4.3.4)						
Endpoints:								
(same as Group B, subgroup 1)								
Subgroup 7			7	7				
Steady state operation life	1027	T_A= 25 °C t=500 hours (see 4.3.4 and figure 5) R_{GK}=1000 ohms I_O= 200 mAdc						

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TABLE III. Group C Inspection (continued)

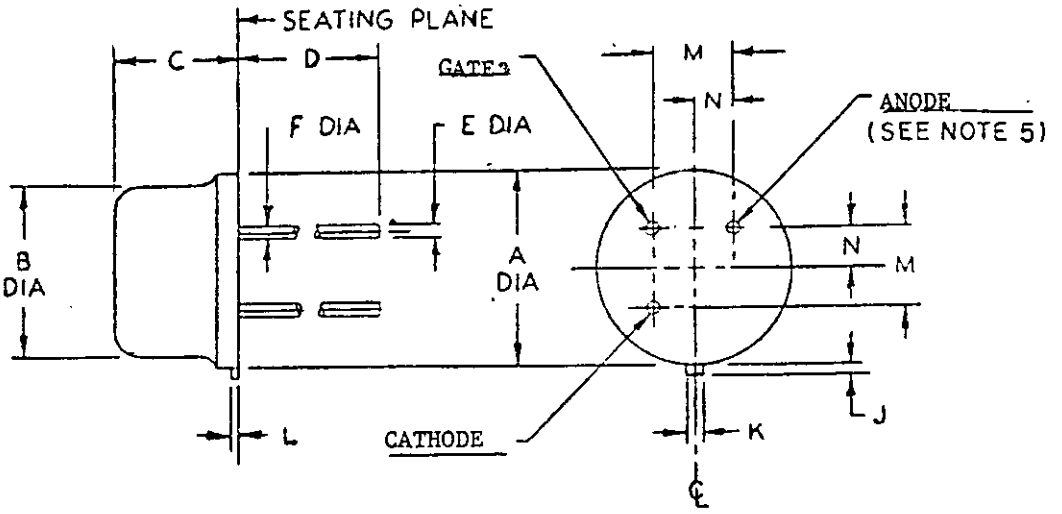
Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX^V		Min	Max	
Subgroup 7 (cont.)								
Steady state operation life (continued)	1027							
2N6605		V_D= 30 Vdc						
2N6606		V_D= 60 Vdc						
2N6607		V_D= 100 Vdc						
2N6608		V_D= 200 Vdc						
Endpoints:								
(same as Group B, subgroup 1)								

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TABLE IV. Burn-In Test Measurements

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TXV TX		Min	Max	
Reverse blocking current	4211	DC method bias cond. D			I_R			
2N6605		V_R = 30Vdc				---	0.25	µAdc
2N6606		V_R = 60 Vdc				---	0.25	µAdc
2N6607		V_R = 100 Vdc				---	0.25	µAdc
2N6608		V_R = 200 Vdc				---	0.25	µAdc
Forward blocking current	4206	DC method bias cond.B			I_{DX}			
2N6605		V_D = 30 Vdc				---	0.12	µAdc
2N6606		V_D = 60 Vdc				---	0.12	µAdc
2N6607		V_D = 100 Vdc				---	0.12	µAdc
2N6608		V_D = 200 Vdc				---	0.12	µAdc
Gate trigger voltage and current	4221	V_D=V_2=6 Vdc						
		R_L=100 ohms						
		R_e =1000 ohms			V_{GT}	---	0.80	Vdc
					I_{GT}	---	200	µAdc

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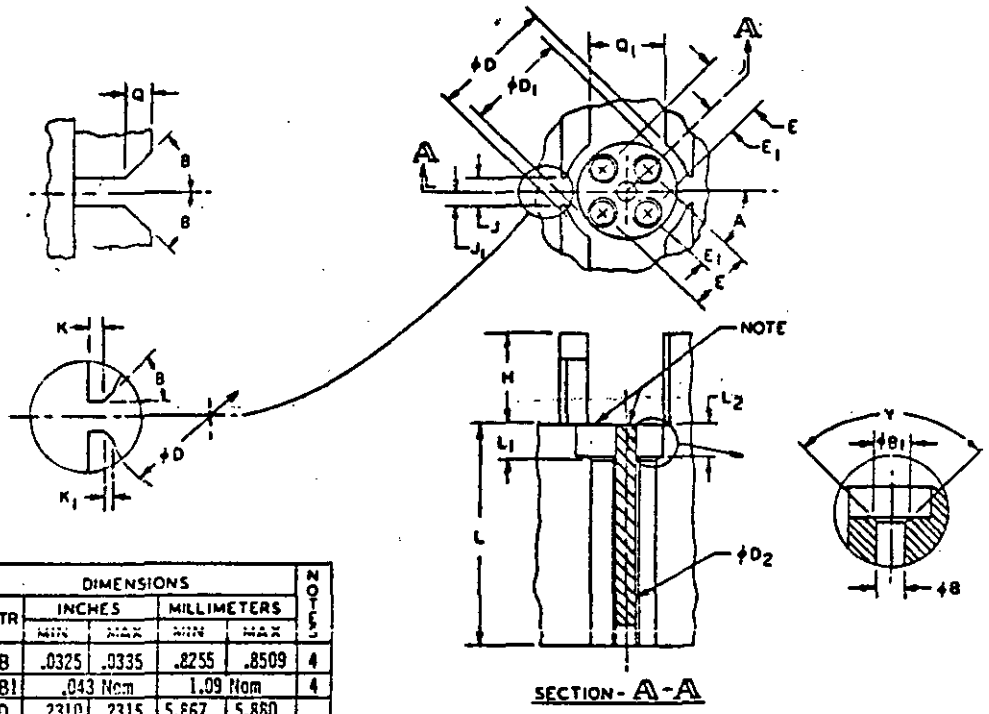


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note 1)		Note
	Minimum	Maximum	
A	.209 (5.31)	.230 (5.84)	
B	.178 (4.52)	.195 (4.95)	
C	.170 (4.32)	.210 (5.33)	
D	.500 (12.70)		7
E		.021 (.53)	2,7
F	.016 (.41)	.019 (.48)	3,7
J	.028 (.71)	.043 (1.22)	6
K	.036 (.91)	.046 (1.17)	
L		.020 (.51)	
M	.0707 (1.80) Nom		4
N	.0354 (.90) Nom		4

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gaging plane .054 +.001,-.000 (1.37 +.03, -.00 mm) below the seating plane of the thyristor, maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 preferred measured method.
5. The anode shall be internally connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All 3 leads.

FIGURE 1. Physical Dimensions of Thyristor Types 2N6605, through 2N6608 (TO-18).

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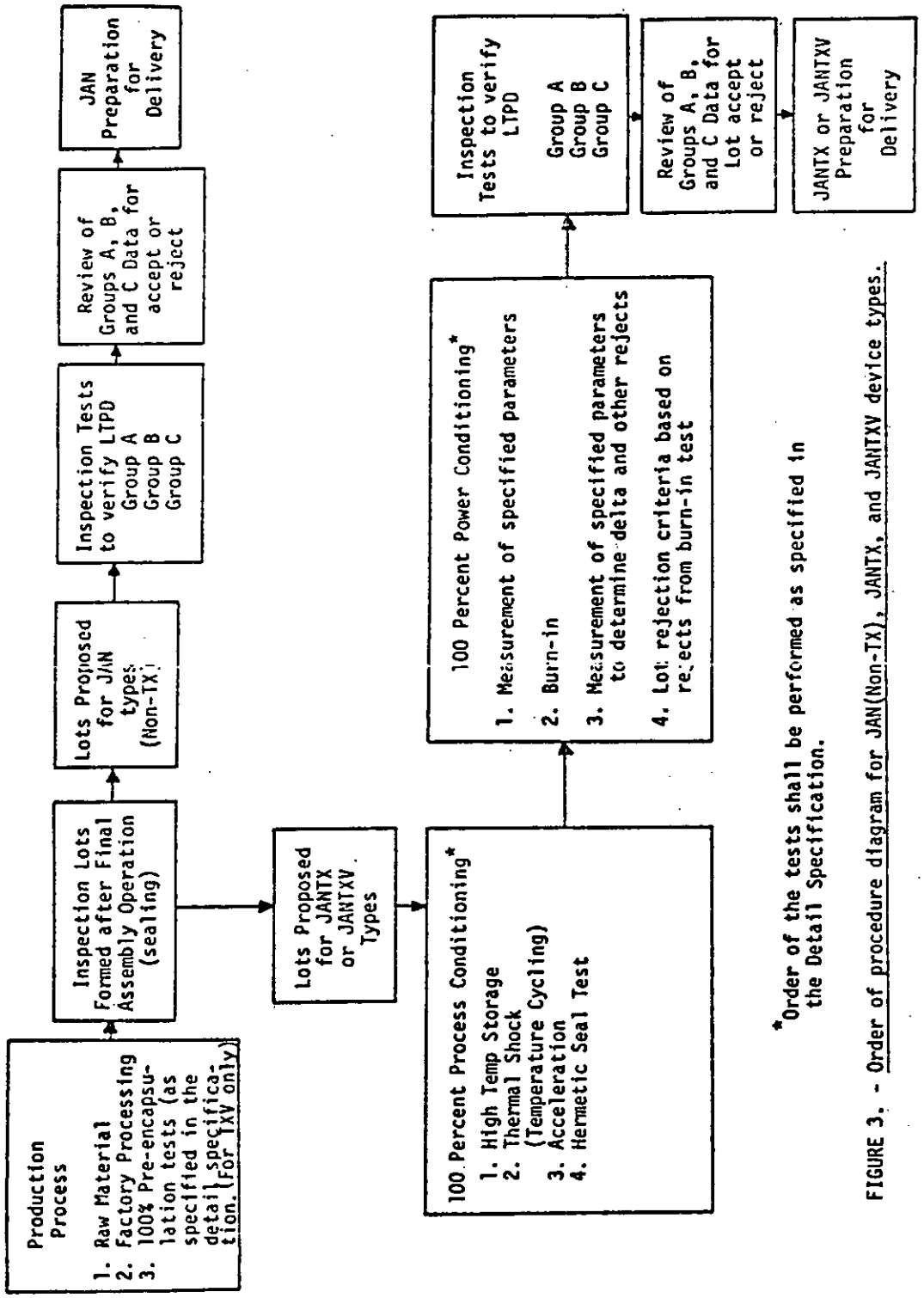


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ØB	.0325	.0335	.8255	.8509	4
ØB1	.043 Nom		1.09 Nom		4
ØD	.2310	.2315	5.867	5.880	
ØD1	.159	.161	4.04	4.09	
ØD2	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.527	2.553	
E1	.0495	.0505	1.257	1.283	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.194	1.207	
J1	.0235	.0245	.597	.622	
K	.009	.011	.229	.279	
K1	.005 Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L1	.054	.055	1.37	1.40	
L2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q1	.123	.127	3.12	3.23	
A	44.90°	45.10°			
B	45° Nom				
Y	90° Nom				

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. The following gaging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 (3.18 mm) + .010 (.254 mm) from the seating surface of the gage. A force of 8±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
   The use of a pin straightener prior to insertion in the gage is permissible. A spacer may be used to obtain the .125 (3.18 mm) distance from the gage seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (.025 mm).
4. Four holes.
5. Pressed in.

FIGURE 2 Gage for lead and tab location for thyristor types 2N6605 through 2N6608.

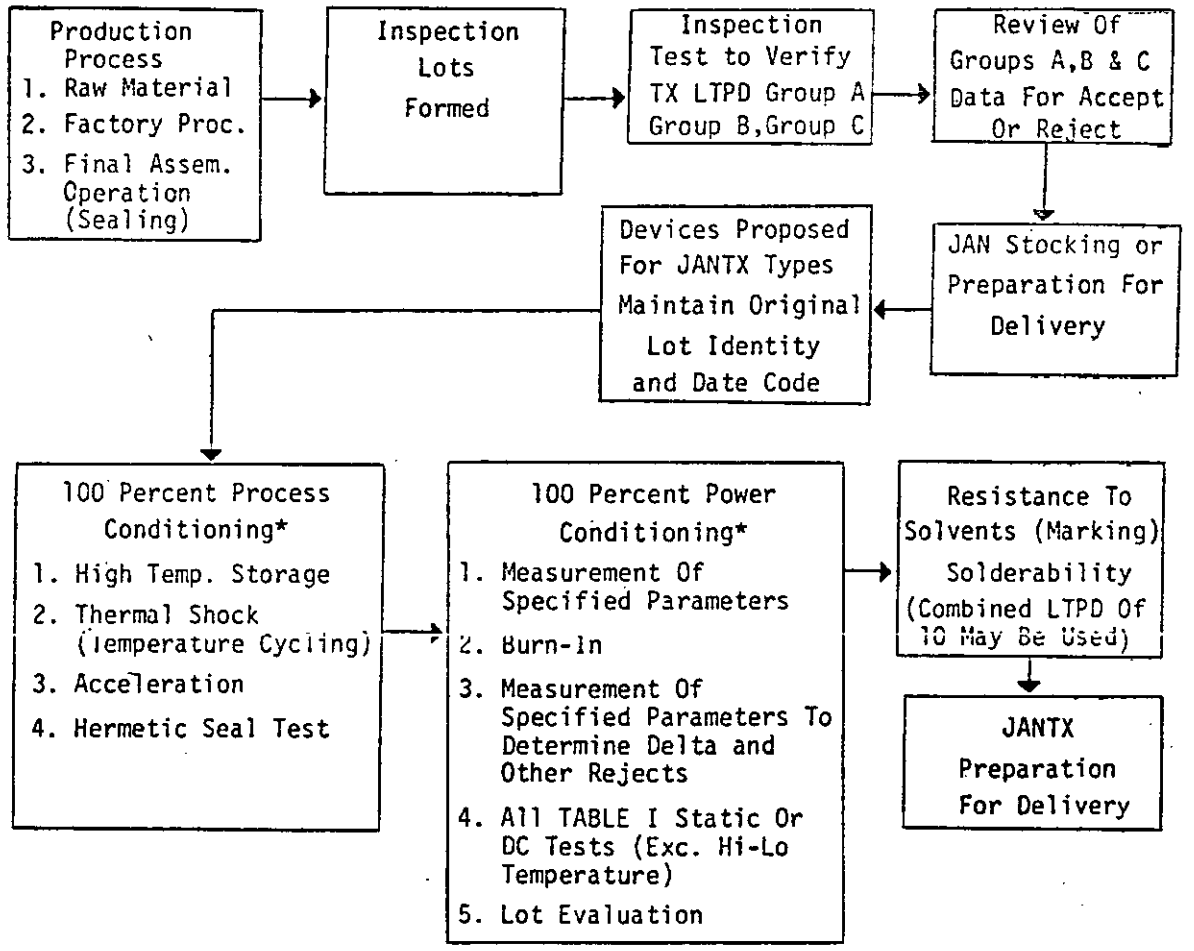
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*Order of the tests shall be performed as specified in the Detail Specification.

FIGURE 3. - Order of procedure diagram for JAN(Non-TX), JANTX, and JANTXV device types.

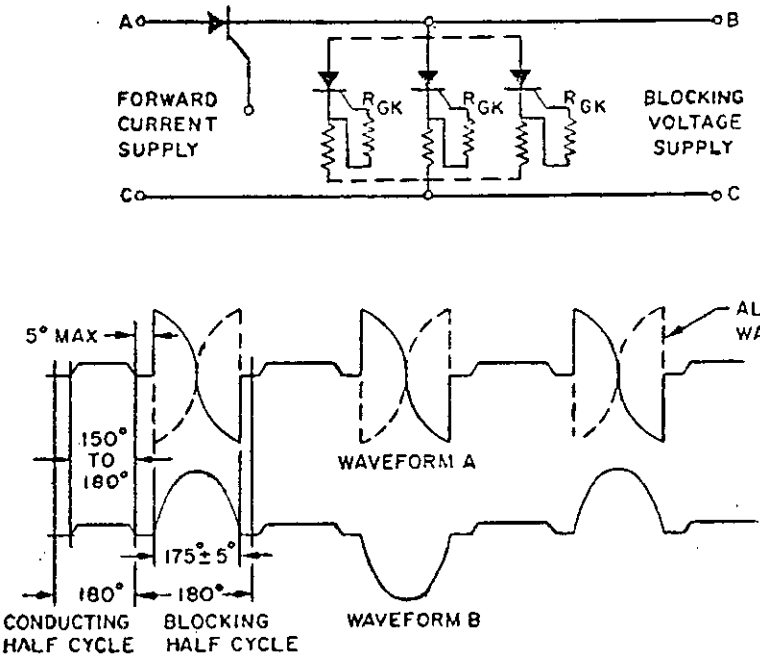
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*ORDER OF THE TESTS IN THE BLOCKS SHALL BE PERFORMED AS SHOWN

FIGURE 4. Alternate order of procedure diagram for JAN (Non-TX), and JANTX device types.

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Test Circuit details: The test circuit used must provide one of the waveforms shown. When a large number of devices are to be tested, they may be connected in parallel or series combinations with suitable provisions (resistive, reactive, etc., methods) for proper division of forward current or blocking voltages. Suitable protective elements should be used to isolate defective devices without interrupting the test and to protect the remaining devices on test.

The forward current supply shall consist of a low-voltage transformer with a means of adjusting the secondary voltage. It shall supply 200mAdc average forward current to each device under test. The gate supply shall be in phase with the forward current supply. The gate average power input shall not be allowed to exceed .01 watt per device.

Operating conditions: The devices shall be subjected to the following operational conditions:

a. Power sources are to be 60 HZ sinusoidal waveform sources.
b. The conduction angle of I_0 shall be 150° to 180°.
c. The rated forward or reverse blocking voltage shall be started during the non-conducting half cycle no later than 5° after conduction has ceased. For waveform A, the RC time constant of the blocking voltage shall be 200+100 microseconds.
d. The duration of the blocking voltages shall be 175±5°

FIGURE 5. Test Circuit and Waveforms for Operation Life Test.

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4.6 Process-Conditioning, Testing, and Screening for "TXV" and "TX" Types. (continued)

4.6.8 Postburn-In Tests. (continued)

4.6.8.1 Burn-In Test Failures (Screening). All devices that exceed the limits of Table IV after burn-in or turn "on" during the burn-in test,shall be removed from the inspection lot and the quantity removed shall be noted on the lot history. If the quantity removed after burn-in should exceed 10 percent of the total inspection lot on the burn-in test, then the entire lot shall be unacceptable as TX or TXV types. If the quantity removed is between 10 and 20 percent the lot may be resubmitted for burn-in, one time only and shall contain only parts which were in the original lot. Resubmitted lots shall be kept separate from new lots and shall be inspected for all the specified characteristics of Table IV and shall not turn "on" during the burn-in test. If a PDA of 5 percent is exceeded for the resubmitted lot, the entire lot shall be unacceptable for TX and TXV types.

4.6.8.2 Electrical Inspection. If the option provided in Figure 4 is used, all devices in the lot must also be submitted to all static or dc tests, except for high and low temperature tests, specified in Table I. All units failing these tests must be removed from the lot and the quantity removed shall be noted on the lot history. The PDA requirements of 4.6.8.1 shall not apply to these parameters.

4.7 Resistance to Solvents and Solderability Tests. If the option provided in Figure 4 is taken, a sample of devices from the lot must be submitted to the tests of Group C, subgroup 2 (resistance to solvent) and Group B, subgroup 3 (solderability only) following 100 percent processing. A sample associated with the LTPD's of the appropriate subgroup or a combined sample for both subgroups using an LTPD of 10 shall be used. These tests are to be conducted one time only on a given lot and shall be conducted on the first sublot of 100 percent processed product. Electrical rejects from the post burn-in inspection may be used for these tests.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, Section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering Data. Procurement documents should specify the following:

(a) Lead material and finish (see 3.3.1).

(b) Inspection data (see 4.3).

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6. NOTES (continued)

6.3 Rating application. The following values may be used as application guidelines:

$+V_{GM}$ = 6V. max.

$+i_{GM}$ = 0.1 a, max.

P_{GM} = 0.1 watt, max.

$P_{G(av)}$ = 0.01 watt, max.

CUSTODIAN:

Army (EL)

PREPARING ACTIVITY:

Army (EL)

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